MECH 325 Machine Design

Group Assignment

Scenario

As a team of students, you are responsible for the safe design of a mechanical system for a client. You are free to select any project that will be of interest to your team. However, it is important that you identify a client for your project. This client may be a student competition team, a friend, or a member of your team. The client can be fictitious if needed. Identifying a client is helpful in establishing customer requirements and evaluation criteria.

For this assignment, we want you to focus on the detailed design phase of the design process. Therefore you should make sure the scope of the project is not too large and fits the criteria for a mechanical design. You are free to make assumptions and estimates of loads in order that you can perform the required calculations. You will use these calculations for sizing components and determining safety factors. Be sure to state the assumptions and the rationale used to make your estimates.

Tasks to be completed

Task 1. Team Formation (required for Group Project and PLC/Hydraulic Labs)

Form a team of 8. Your team can consist of team members from either section of the course. You are welcome to form teams of less than 8 students that will be then be populated with other students to make up the minimum requirement of team members. If you wish to have a team of 6, 7, or 9 students, please contact me in advance.

The following options will be available for you to consider:

1. You have formed a complete team. **ONE student** enter the information into the Canvas Assignment. The information will include the names of your team members. If you have already determined a topic, you are welcome to enter the title of the topic. Again, only ONE student from your team needs to enter this information.
2. You have formed a small group of less than eight students and wish to recruit additional students to your team. You have a wonderful idea for a project that may be of interest to other students. Use the Discussion Forum to assist with recruiting additional students to your team. Even if you have less than the required number of students on your team, be sure to enter the information on the Canvas Assignment so your team can remain together (**only one** of you needs to enter the information). Following the deadline, other students will be added to make the team complete.
3. As an individual you have a great idea for a project that will interest fellow students. Use the Discussion Forum to recruit group members. If you find additional team members, be sure to

have one member of the team enter the information on Canvas (even if the group size if less than the required number).

4. Wait and you will be placed into a group.

   Submission Deadline: **Monday, Sept. 14th**. Final composition of teams will occur soon after Sept 14th. Please note that the course instructor reserves the right to alter teams.

Project Proposal.

Task 2. Propose a project for the team.

As stated earlier the project needs to focus on a mechanical design. This will allow you to apply the principles and topics that you are being taught in this course. These include flexible drives, wire ropes, hydraulic systems, shafts, gears, fasteners, bearings and springs. You are not required to include all the elements in your design but you should aim for a minimum of two elements. You will likely need to consider additional elements in your design such as motor selection, frame sizing, etc. However, you are not required to perform detailed calculations on these items.

Ideas for projects include:

Example Design Projects from 2023 to 2025. See the list at the end of the document. Design topics from other years include: Epicyclic Gearbox Analysis for Formula Electric Car, Hydraulic Gimbal Actuator for UBC Rocket, Drivetrain Optimization for SUBC Vessel, Rudder Actuation Drive for Sailbot, and a Bike Lift Storage System. It must be noted that a large portion of the projects are in support of the various student competition teams.

Other ideas for topics can be found in Chapter 23 of the Mott textbook. There are some great design projects that you can modify to better fit your interests. Some interesting ideas include amusement rides for children, and various lifting devices.

Once you have decided on the project, prepare a memo outlining your proposed project. In the memo you should identify the objective of your project, and the client. If the overall project is large in scope (perhaps the powertrain for a vehicle), identify the portion of the project your team is responsible to complete (say the rear differential only). The memo will be reviewed to ensure the project meets the requirements of the course.

Submission Deadline: Midnight, Friday, Oct 9th

Report

Task 3. Complete the design and required analysis and prepare a summary report.

Your engineering report will highlight all the key findings of your project. As a general guideline for the report, you should address the categories outlined in the evaluation form that will be used for your presentation on December 3rd. It is expected that the written report will be approximately 8- 10 pages in length including Tables and Figures (not including Appendices). The report should outline all the key results from your analysis leading to your conclusions and recommendations. Details of the analysis should be included in the Appendices. This includes sample calculations, spreadsheet output, simulations, catalogue pages, detailed drawings, etc. Appendices for this type of project ranges from 10 to 30 pages in length.

It is recommended that you review Mott Chapter 1 especially Sections 1-4 to 1-7. This chapter provides a good summary on the design process for mechanical components.

Remember that the emphasis behind this project is the application of the principles learned in this course. You are welcome to use software to aid with analysis but you should expect to provide a sample calculation to validate the output. You are encouraged to use CAD software (Solidworks, AutoCAD, etc.) to aid with the design. However, a detailed CAD model of your design is not a specific requirement for the assignment.

Task 3 Report Deadline: Midnight Monday, Dec. 7th

Presentation

Task 4 Present your design for evaluation.

Your team will be required to present the results of your group project to your fellow classmates, TA's, instructors, and guests. We will run simultaneous sessions during the lecture and tutorial sessions on Dec. 3rd. Each team will be allocated 20 minutes to present their design and answer questions. It is recommended that the presentation be 15 minutes to allow 5 minutes for questions and answers.

Each session will have 4 or 5 teams and you are required to evaluate all the teams in your session. As a team, you will agree on the scores awarded to the team you are evaluating (this may be an average of all the individual scores on your team, or you may discuss the results after the session to agree on a score). The evaluation form is attached.

Task 4 Deadline: Midnight Thursday, Dec. 3rd

Further information and discussions will take place during class and tutorial sessions.

Team Being Reviewed:_____ Reviewer:_____

MECH 325 Mechanical Design Project

1. Intro and recap of main project
   Why did your team choose to work on this project? Is it part of a larger project such as a completion vehicle? Have you identified a client?

2. Purpose of the mechanical project explained clearly
   What are the design requirements that you identified for this project? Are you able to identify some key evaluation criteria? What are the key components that you are analyzing?

3. Appropriateness of the detailed analysis

   Did you make assumptions and/or simplifications? If so explain them. What inputs did you use? What types of factors did you use for your analysis (overload/service factors, reliability, design factor, etc.)?

4. Results of the analysis
   What did you determine (max. loads/stresses, power required, speeds, costs, etc)? Have you sized the right components for your design? Were you able to establish the factor of safety for your design analysis (static loading, fatigue, etc.)?

5. Conclusions and/or Recommendations

Do you feel confident in the results of your analysis? Did you meet the design requirements and pass your evaluation criteria? Is your design safe? Do you feel the results warrant the design to be implemented (for example, build a prototype)? Do you have recommendations for further work (detailed design phase, FEA, etc.)?

6. Other comments

Simulation output, detailed graphics, professional presentation, etc.

Grade /10

Presentation Schedule and Project Titles from 2025:

MECH 325 Group Project Presentation Schedule
Thursday, Dec. 4th

WOOD 1
9:30-1100

Evaluator	Team	
Jon	13	Design of a Drivetrain for a Single-speed Bicycle
Syed	26	Design a Modified Chainsaw for UBC Thunderjacks
Zahra	16	Design of an Automatic Baseball Launcher
Omid	23	Design of a Constrcution Lift

LIFE 2402
9:30-11:00

Evaluator	Team	
Giovanni	1	Design of a Pedal Boat Drive System
Yasin	9	Design of an Adjustable Steering Column for Formula UBC
Desmond	20	Log Splitter Power Train Design
Nima	28	Design of a Drivetrain Steering System for a Go-kart

WOOD 1
11:00-12:30

Evaluator	Team	
Jon	12	Design of a Gearbox for UBC Rover Team
Syed	6	Design of a Hand-cranked Communal Fishing Winch
Zahra	10	Design of an Incline Treadmill
Omid	5	Human Powered Vehicle Drivetrain

LIFE 2402
11:00-12:30

Evaluator	Team	
Giovanni	7	Design of an Equatorial Camera Mount
Yasin	4	Design of a Articulated Robot Arm for Radioactive Medical Supplie
Desmond	29	Design a Centrifuge for Home Brewing
Nima	32	Design of a Robot Arm for Muscular Dystrophy Canada

High Dempster 101
2:00-3:30

Evaluator	Team	
Jon	2	Moped Drivetrain System Design
Fardad	11	Design of a 3 Degree of Freedom Robotic Arm
Omid	21	Design of a Height-Adjustable Desk
Nima	3	Portable Rope Tow for Backcountry Skiing

FNH 30 (Food & Nutritional Health)
2:00-3:30

Evaluator	Team	
Yuecheng	15	Design of a Tray-handling System for Orchard Commons' Kitchen
Atefeh	8	Design of a Scissor Lift Mechanism
Raymond	18	Design of a Mechanical System for a Recreational Crossbow
Justin	25	Design of a Carousel System for Storing Rolls of Materials

High Dempster 101
3:30-5:00

Evaluator	Team	
Jon	17	Design of a Portable Sawmill
Fardad	31	Design of a Tennis Ball Launcher
Omid	27	Design of a Contra-rotating Gearbox and Shaft for SUBC
Syed	14	Design of an Antenna Tracking System for UBC Rocket

FNH 30 (Food & Nutritional Health)
3:30-5:00

Evaluator	Team	
Yuecheng	30	Design of an Automatic Mango Sorting/grading Machine
Atefeh	24	Design of a Hand-Powered Coffee Grinder
Raymond	22	Design for an Electric Bicycle
Justin	19	Design of a Leveling System for an Outer Space Agriculture System

Presentation Schedule and Project Titles from 2024:

MECH 325 Group Project Presentation Schedule
Thursday, Dec. 5th

WOOD 5
9:30-1100

Evaluator	Team	
Jon	10	Regenerative Braking System for Bicycles
Syed	15	Design of a Windlass System for Ships
Zahra	22	Limited-Slip Differential for Formula UBC Vehicle
Omid	28	Design of a 3D Printer Gantry for UBC Rocket
Zareer		

CEME 1203
9:30-11:00

Evaluator	Team	
Rahmi Can	21	Cable Arresting Mechanism for Landing Aircraft
Yasin	12	Design of a Drivetrain for a Quadracycle
Giovanni	2	Design of an Open Differential for a 1990 Corvette
Kyle		
Mobin		

WOOD 5
11:00-12:30

Evaluator	Team	
Jon	7	Drivetrain System for UBC Supermileage
Yuecheng	13	Gearbox Design for UBC Formula Electric
Zahra	26	Design of a Baseball Pitching Machine
Omid	25	Thrust Vectoring System for UBC Rocket
Syed		

Scarfe 1021
11:00-12:30

Evaluator	Team	
Rahmi Can	8	Swimming Pool Lane Rope Reeling Device
Yasin	19	Active System for Aero Rear Wing on UBC Formula E
Giovanni	14	Design of a Prosthetic Leg
Kyle	6	Design of Mini Golf Course Obstacles

CEME 1203
2:00-3:30

Evaluator	Team	
Jon	3	Design of a Drivetrain for a Drift Trike
Mobin	24	Paddleboard Outboard Propulsion Attachment System
Zareer	4	Design of a Selective Compliance Articulated Robot Arm ('
Omid	30	Design of a Grinder Attachment for a Lathe
Seyed		

FNH 30 (Food & Nutritional Health)
2:00-3:30

Evaluator	Team	
Maryam	16	Design of a Food Serving Conveyor Belt
Yuecheng	20	Soft-Stop Mechanism for Food Tray Coveyor Belt
Atefeh	31	Design of a Suspension System for Monster Truck
Zahra	23	Design of a Drill System for a Mars Rover
Rahmi Can		

CEME 1203
3:30-5:00

Evaluator	Team	
Jon	27	Design of a Penny Press Machine
Mobin	9	Design of a Device to Assist Mural Artists
Zareer	5	Design of a Vending Machine
Omid	32	Design of a 3D Printer Bed
Seyed		

FNH 30
3:30-5:00

Evaluator	Team	
Maryam	1	Design of a Hydraulic Ant-Roll Bar System
Rahmi Can	11	Design of an XY Table for 3D Printer
Atefeh	17	Design of a Trebuchet
Zahra	18	Drivetrain for a Hybrid Bicycle